

Pointed Involutions, KEP 30 | 14 Normal Forms, and the Roof–Rail Reflection Frontier

A proof-first KEP/RH/Stone refinement of the 44-split and the $89 = 44 + 45$ closure

Formal working note

27 August 2026

Abstract

We refine the roof–rail recurrence comparison by introducing a finite C_2 -set invariant and by replaying source-bound Kepler (KEP), RH/A1BE, and Stone carriers. An exact A1BE pointed 89-element involution is constructed from the 44 complete even reversal two-cycles together with the distinguished fixed point (3, 26). It has one fixed point, 44 free two-cycles, and 45 total orbits; consequently

$$89 = 44 + 45$$

is realized structurally, not merely arithmetically.

KEP-R13 supplies the native decomposition

$$44 = 9 + 21 + 7 + 7, \quad 30_{\text{even}} = 9 + 21, \quad 14_{\text{odd}} = 7 + 7.$$

KEP-R27 supplies the exact Cube27 atomic decomposition

$$27 = 1 + 12 + 6 + 8 = (1 + 12) + (6 + 8) = 13 + 14.$$

These data give canonical transfer sizes 21, 13, 12 for the three recurrence-positive roof branches $9 | 35$, $17 | 27$, and $18 | 26$. The negative branch $8 | 36$ would require transfer size 22, which is not a union of the declared R13 face classes nor of the declared R27 atomic blocks. This is a scoped no-go only: the independent Stone width-three reversal has an exact 22-state partial-action domain, with escape labels 8, 17.

The RH rectangle of 36 points has six fixed points and fifteen two-cycles; its involution split is $36 = 15 + 21$, supplying an independent structural realization of the positive mover 21. The remaining roof/KEP/RH-to-recurrence commuting square is left explicitly SEPARATED/NT.

1 Finite involutions and the canonical orbit split

Definition 1.1. *Let X be a finite set equipped with an involution J . Write*

$$f = |\text{Fix}(J)|, \quad p = \#\{\text{free two-cycles of } J\}, \quad q = |X/J|.$$

Theorem 1.2 (Involution split). *For every finite involution,*

$$|X| = 2p + f, \quad q = p + f,$$

and therefore

$$\boxed{|X| = p + q.}$$

Moreover $q - p = f$.

Proof. The J -orbits are precisely f singleton fixed orbits and p two-element free orbits. Hence $|X| = f + 2p$ and $q = f + p$. $\square$

Corollary 1.3 (Pointed case). *If J has a unique fixed point, then*

$$|X| = 2p + 1, \quad |X/J| = p + 1,$$

so

$$\boxed{|X| = p + (p + 1).}$$

2 An exact RH/A1BE pointed 89-carrier

Let E be the 212-element even support in the frozen A1BE source and let

$$J(i, j) = (29 - j, 29 - i).$$

The source replay has exactly seven fixed support pairs and 44 complete two-cycles. Let T_{88} be the union of the 88 points in those 44 complete two-cycles and let

$$p_\star = (3, 26)$$

be the F1-distinguished fixed pair. Define

$$P_{89} = T_{88} \sqcup \{p_\star\}.$$

Theorem 2.1 (Source-bound pointed 89-involution). *The set P_{89} is J -stable, has cardinality 89, and has exactly one fixed point. Consequently*

$$\boxed{\#\{\text{free } J\text{-orbits}\} = 44, \quad |P_{89}/J| = 45,}$$

and hence

$$\boxed{89 = 44 + 45}$$

as the free-orbit count plus the total-orbit count of one explicit source-bound involution.

Proof. The 44 complete two-cycles contribute 88 pairwise distinct points and are J -stable. The distinguished point $p_\star$ is fixed and is not in a two-cycle. Thus P_{89} has one fixed orbit and 44 free orbits. Apply the pointed involution corollary. $\square$

Guard 2.2. *This theorem does not identify the A1BE P_{89} carrier with the prime 89, with the Fibonacci carrier F_{11} , or with the order-89 recurrence carrier. It realizes only the same integer decomposition $89 = 44 + 45$ by a genuine involutive structure.*

3 The RH rectangle 36 and a native 15 | 21 split

The F1 rectangle component has $36 = 6 \cdot 6$ points and is closed under the same reversal. Its exact orbit census is

$$f = 6, \quad p = 15.$$

Hence

$$q = p + f = 21$$

and

$$\boxed{36 = 15 + 21.}$$

Equivalently, for the permutation representation,

$$\dim V_+ = 21, \quad \dim V_- = 15.$$

The number 21 therefore occurs as an RH-native quotient/even dimension, independently of the KEP $F4opp$ count used below.

4 KEP-R13: a native $9 \mid 35 \rightarrow 30 \mid 14$ normal form

The recovered R13 endpoint carrier has four face classes:

$$|F6| = 9, \quad |F4_{opp}| = 21, \quad |F5_{opp}| = 7, \quad |F5_{adj}| = 7.$$

Grouping by active-edge parity yields

$$\boxed{30_{\text{even}} = 9 + 21, \quad 14_{\text{odd}} = 7 + 7.}$$

Therefore the split

$$9 \mid 35 = 9 \mid (21 + 7 + 7)$$

has the native KEP transfer

$$\boxed{9 \mid 35 \xrightarrow{F4_{opp}21} (9 + 21) \mid (7 + 7) = 30 \mid 14.}$$

This is a genuine decomposition of one R13 object carrier, not merely a comparison of dimensions.

5 KEP-R27: Cube27 transfer blocks

Inside the 27-point carrier $\{-1, 0, 1\}^3$, the declared atomic classes have cardinalities

$$\boxed{1 \quad (\text{centre}), \quad 12 \quad (\text{cuboctahedral roots}), \quad 6 \quad (\text{axes}), \quad 8 \quad (\text{cube corners}).}$$

The parity split is

$$\boxed{C_{13} = 1 + 12, \quad O_{14} = 6 + 8, \quad 27 = 13 + 14.}$$

Consequently, after explicitly declaring a tagged right-hand Cube27 carrier,

$$\boxed{17 \mid 27 = 17 \mid (13 + 14) \longrightarrow (17 + 13) \mid 14 = 30 \mid 14.}$$

Likewise, on the punctured Cube27 carrier

$$26 = 12 + 6 + 8 = 12 + 14,$$

so

$$\boxed{18 \mid 26 = 18 \mid (12 + 14) \longrightarrow (18 + 12) \mid 14 = 30 \mid 14.}$$

Guard 5.1. *Unlike the R13 $9 \mid 35$ transfer, the preceding two displays are typed tagged-carrier normal forms. They use native KEP subcarriers on the right, but do not assert that the left 17- or 18-object roof/recurrent carrier is literally a KEP subset.*

6 The scoped 22-mover obstruction

For a split $a \mid (44 - a)$, transfer to $30 \mid 14$ by moving a right-hand block to the left requires a block of size

$$m(a) = 30 - a.$$

For the four roof-selected branches:

split	$m(a)$	recurrence reflection
$8 \mid 36$	22	no
$9 \mid 35$	21	yes
$17 \mid 27$	13	yes
$18 \mid 26$	12	yes

Let $\mathcal{M}_{R13}$ be the subset-sum set generated by 9, 21, 7, 7, and $\mathcal{M}_{R27}$ the subset-sum set generated by 1, 12, 6, 8. Then

$$21 \in \mathcal{M}_{R13}, \quad 13, 12 \in \mathcal{M}_{R27},$$

but

$$22 \notin \mathcal{M}_{R13} \cup \mathcal{M}_{R27}.$$

Thus the three recurrence-positive roof branches have a transfer by one of the declared canonical KEP block unions, whereas $8 \mid 36$ does not.

Guard 6.1 (Finite alignment, not a general classifier). *This agreement is proved only on the four roof-selected branches. The KEP subset-sum criterion does not reproduce the complete 21-pair recurrence classification at $k = 44$, and it is not promoted to a causal explanation of recurrence reflection.*

7 Stone supplies the missing cardinal 22, but on another carrier

Encode $r \in \{0, \dots, 23\}$ as a width-three ternary word and reverse its digits. The closed partial-action domain is

$$D_{22} = \{0, \dots, 23\} \setminus \{8, 17\}.$$

Indeed

$$8 = 022_3 \mapsto 220_3 = 24, \quad 17 = 122_3 \mapsto 221_3 = 25.$$

Thus $|D_{22}| = 22$. On D_{22} , reversal has eight fixed points and seven free two-cycles. On the minimal total closure $\{0, \dots, 25\}$, it has eight fixed points and nine two-cycles, so the involution split gives

$$26 = 9 + 17.$$

Therefore 22 is not globally absent. The precise current statement is:

$$22 \text{ is absent from the declared canonical KEP mover class,}$$

while an independent Stone 22-carrier exists and has the rail escape labels 8, 17. Constructing a structure-preserving map from this Stone carrier into the $8 \mid 36$ normal-form problem is a new, sharply typed target.

8 F1 Fourier 9/21 and the correct categorical level

On the 6×6 Fourier grid, independent antipody invariance leaves nine modes. Under the swap involution $J(p, q) = (q, p)$, the J -even sector has

$$6 + 15 = 21$$

modes and the J -odd sector has 15 modes. Hence the 9/21/15 profile is exact.

Later KEP continuation constructs deterministic $9 \rightarrow F6$ and $21 \rightarrow F4opp$ representative maps and then upgrades them to orbit-valued functors. The native orbit saturations have sizes 60 and 42, so a direct equivariant bijection of the original unsaturated sets is impossible. This identifies the appropriate future category: orbit/groupoid-valued transport rather than raw finite-set identity.

9 A prediction for $34 \mid 55$

For an 89-point involution,

$$p = \frac{89 - f}{2}, \quad q = \frac{89 + f}{2}.$$

Therefore

$$p = 34, \quad q = 55 \iff f = 21.$$

Thus a structural realization of

$$89 = 34 + 55$$

by the same involution-split mechanism would require a native 89-point C_2 -carrier with exactly 21 fixed points.

No such source-bound carrier is established in the present replay. The statement is therefore an NT target, not a theorem of existence.

10 Remaining recurrence commuting square

The recurrence reflection acts on roots by

$$\rho_{\text{rec}}(\alpha) = -\alpha^{-1}.$$

The next proof target is an explicit typed map Ψ from one of the source involution/groupoid carriers above into the recurrence factor-orbit carrier such that

$$\Psi \circ J_{\text{source}} = \rho_{\text{rec}} \circ \Psi.$$

At present:

Statement	Status
A1BE pointed P_{89} : 44 free cycles and 45 total orbits	PASS
Structural involution identity $89 = 44 + 45$	PASS
RH rectangle $36 = 15 + 21$	PASS
Native KEP $9 \mid 35 \rightarrow 30 \mid 14$	PASS
Tagged Cube27 $17 \mid 27 \rightarrow 30 \mid 14$ via 13	PASS typed
Tagged punctured-Cube27 $18 \mid 26 \rightarrow 30 \mid 14$ via 12	PASS typed
Canonical KEP block-union mover 22 for $8 \mid 36$	REFUTED scoped
Stone 22-domain with escape labels 8, 17	PASS separate
Native 89-point involution with 21 fixed points	NT
Full source-to-recurrence commuting square	SEPARATED/NT

Source locks

The replay used the following source SHA-256 values:

- A1BE_direct_tail_pairs.jsonl:
7fa2ac47faa32aa5170da5e18bb4fff236239b9d52f55021292b75a3e907e881;
- A1BE_F1_reversal_analytic_morphism.json:
b3ff6920ef9b1303892f5ced198fc1f33154194682891a9856bb545ea436a8c6;
- II_I0_KEP_R13_stable44_ledger.json:
62b9b76f3aec2a4e6102f83be144cca94ba25e1856885e2d9278c77b422843df;
- kep_d3_fcc_box_transport_formalization_R27.tex:
1a95d8f8c4dcdfdf16a6c605d1effa40b0ab4b7c17a201290212d1de0a1312c4;
- II_HW_partialgroupoid_zRquotient_ReesAutomaton_H3cocycle_R1.tex:
449c3e82733abe02fa2a439867865b17cbc03cc8e91bab66bc9f0deb12138fe6.